CURRICULUM VITAE

ALLISON WELCH

EDUCATION

2025 **University of California, Irvine,** CA
Ph.D. in Earth System Science
Advisor: Dr. Claudia Czimczik
Dissertation: Changing Arctic Vegetation: Examining Shrubification, Permafrost Vulnerability, and Land-Atmosphere Interactions

2022 **University of California, Irvine**, CA
M.S. in Earth System Science

2020 **University of Florida,** Gainesville, FL
B.Sc. in Environmental Science, Summa cum Laude
Thesis: Evaluation of Element/Ca Ratios as a Screening Technique to Quantify the Amount of Diagenetic Calcite in Fossil Corals
Minor: Geology; Wildlife Ecology & Conservation
Certificate: College of Agriculture and Life Sciences Honors Scholar

EXPERIENCE

SEP 2025–PRESENT **Postdoctoral Research Fellow –** University of Alaska, Fairbanks

- Organized, executed, and led a time-sensitive field campaign to capture aerial images of high latitude lakes during freeze up (Fall 2025) using drones (Fairbanks and Toolik Lake, AK).
- Independently developing methodology for geospatial analysis at the regional scale.
- Using aerial images of frozen methane bubbles in high latitude lakes and machine learning to improve Arctic carbon balance estimates.

JUN 2025–AUG 2025 **Lecturer –** UCI Summer Session

- Taught Introduction to Earth System Science (General Education requirement) to a class of 70 students
- Developed and presented 26 lectures, 9 quizzes, 9 discussion activities, and 2 exams with content related to the fundamentals of Earth science and climate change

OCT 2021–JUN 2025 **Teaching Assistant** – UCI Department of Earth System Science

- Taught upper- and lower-level courses in field methods, physics, and sustainability for 1000+ hours and >500 students in lecture, discussion, and laboratory settings
- Led 150+ students in group projects in sustainability and geophysics by providing individualized feedback and teaching writing fundamentals to improve their understanding of the scientific method
- Devised lesson plans and worksheets using the CREATE (Consider, Read, Elucidate the hypotheses, Analyze and interpret the data, Think of the next Experiment) framework

OCT 2020–JUN 2025	**Graduate Student Researcher** – UCI Department of Earth System Science • Performed data collection and analysis by combining field measurements (vegetation surveying, aerosol sampling, sample collection), laboratory measurements (radiocarbon and elemental analysis), remote sensing (GIS), and modeling to produce integrated results • Synthesized results and literature reviews to create papers for peer-reviewed publication
JUL 2024–DEC 2024	**Research Assistant Intern** – Woodwell Climate Research Center • Parameterized and calibrated the Geophysical Institute Permafrost Model (GIPL) using monitoring data • Used field observations to model ground temperatures and permafrost conditions under scenarios of warming, snow addition, and vegetation change • Successfully obtained NSF Funding (INTERN Grant, ~$50,000) to support collaboration with Woodwell
MAY 2019–MAR 2020	**Volunteer Research Assistant** – U.S. Geological Survey • Performed data collection (filtered eDNA water samples for DNA isolation and extraction), and analysis (PCR, qPCR) for invasive species research in conservation genetics
MAY 2018–AUG 2018	**Land Management Intern** – Alachua Conservation Trust • Performed manual and chemical invasive flora species removal, wildlife/botanical surveys, general upkeep to improve visitation to public lands • Continued to work after completion of internship to lead environmental education events about wildlife and prescribed burning for the public
NOV 2017–MAY 2020	**OPS Undergraduate Research Assistant** – UF Department of Geological Science • Cut, catalogued, and analyzed fossil coral samples used in studies of paleoclimatic sea level changes • Synthesized radiocarbon data from literature • Gathered x-ray diffraction (mineralogy), ICP-MS (elemental composition) data and analyze using R to assess patterns in geochemical alteration of coral skeletons

LEADERSHIP & INVOLVEMENT

2024	**Board Member**, Department of Earth System Science Graduate Leadership
2023–2024	**Seminar Lead**, Department of Earth System Science Seminar Series; Half-Baked Seminar Series
2023	**Writer, Administrator, and Grader,** Orange County Science Olympiad (Division C)
2023–	**Mentor**, Earth System Science Undergraduate Research Students: Nicole Nielsen
2023–	**Educator**, Climate, Literacy, Education, And iNquiry (CLEAN) Initiative
2022–2024	**Mentor and Judge**, Irvine Unified School District Science Fair, Orange County Science and Engineering Fair

FIELD CAMPAIGNS

JUL–AUG 2023	Toolik Field Station
JUL–AUG 2022	Toolik Field Station
MAR 2022	Blodget Forest Research Station

COURSES TAUGHT (TEACHING ASSISTANT)

SP 2025	Environmental Sustainability II
WI 2025	Environmental Sustainability I
SU 2024	Intro to Earth System Science
SP 2024	Earth System Physics
WI 2024	Food and Water Systems
SP 2022	Environmental Sustainability II
WI 2022	Environmental Sustainability I
FA 2021	Field Methods

FELLOWSHIP

2024	National Science Foundation Non-Academic Research Internships for Graduate Students (INTERN) (~$50,000)
2023	Ridge 2 Reef Graduate Fellowship (NSF DGE 1735040) ($34,000)
2017–2020	Doris Lowe and Earl and Verna Lowe Scholarship ($1500/year)
2017	University of Florida Summer Merit Study Abroad, New Zealand ($1000)
2016–2020	Florida Academic Scholars ($6381/year)

AWARDS

2020	UF College of Agriculture and Life Sciences (CALS) Honors Award
2017–2020	Society of Phi Kappa Phi
2017–2020	Anderson Scholar
2016–2020	UF Honors Program
2016–2020	UF Presidents Honor Roll
2016–2020	UF CALS Dean's List
2016	International Baccalaureate Diploma Recipient

PUBLICATIONS

Journal Articles

Welch, A. M., Matthews, T., Sheesley, R. J., Wang, H., Barsanti, K. C., Nielsen, N., Xu, X., Nui, L., Guenther, A. B., & Czimczik, C. I. (2025). Summertime carbonaceous aerosol in interior vs. coastal Northern Alaska. *Journal of Geophysical Research: Atmospheres 130*(8) e2024JD042080. https://doi.org/10.1029/2024JD042080

Djokic, M., Hill, R., Anderson, A., Barbour, K., Coleman, A., Guerra, A., Hunt, C., Jolly, A., Long, J., Manley, K., Montoya, J., Norlen, C., Nugent-Suratt, A., Washburn, K., Weber, S., **Welch, A**., Wong, C., & Allison, S. (2024). From Ridge 2 Reef: An interdisciplinary model for training the next generation of environmental problem solvers. *PLOS ONE, 19*(12), e0314755. https://doi.org/10.1371/journal.pone.0314755

Wang, H., Nagalingam, S., **Welch, A. M.,** Leong, C., Czimczik, C. I., & Guenther, A. B. (2024). Heat waves may trigger unexpected surge in aerosol and ozone precursor emissions from sedges in urban landscapes. *Proceedings of the National Academy of Sciences, 121*(45), e2412817121. https://doi.org/10.1073/pnas.2412817121

Wang, H., **Welch, A. M.,** Nagalingam, S., Leong, C., Kittitananuvong, P., Barsanti, K. C., ... & Guenther, A. B. (2024). Arctic heatwaves could significantly influence the isoprene emissions from shrubs. *Geophysical Research Letters, 51*(2), e2023GL107599. https://doi.org/10.1029/2023GL107599

Wang, H., **Welch, A. M.,** Nagalingam, S., Leong, C., Czimczik, C. I., & Guenther, A. B. (2024). High temperature sensitivity of Arctic isoprene emissions explained by sedges. *Nature Communications, 15*, 6144. https://doi.org/10.1038/s41467-024-49960-0

Welch, A. M., Pedron, S. A., Jespersen, R. G., Xu, X., Martinez, B., Khazindar, Y., Fiore, N. M., Goulden, M. L., & Czimczik, C. I. (2023). Implications of alder shrub growth for alpine tundra soil properties in Interior Alaska. *Arctic, Antarctic, and Alpine Research, 55*(1). https://doi.org/10.1080/15230430.2023.2285334

Yanez, C. C., Hopkins, F. M., Xu, X., Tavares, J. F., **Welch, A.,** & Czimczik, C. I. (2022). Reductions in California's urban fossil fuel CO_2 emissions during the COVID-19 pandemic. *AGU Advances, 3*(6), e2022AV000732. https://doi.org/10.1029/2022AV000732

Datasets

Welch, A. M., Matthews, T., Sheesley, R. J., Wang, H., Barsanti, K. C., Nielsen, N., Xu, X., Nui, L., Guenther, A. B., & Czimczik, C. I. (2024). *Characterization of summertime carbonaceous aerosol on the North Slope of Alaska (Toolik and Utqiaġvik), 2022–2023* [Dataset]. Arctic Data Center. https://doi.org/10.18739/A2V40K16P

Pedron, S., **Welch, A. M.,** Jespersen, R. G., Xu, X., Martinez, B., Khazindar, Y., Fiore, N., Goulden, M. L., & Czimczik, C. I. (2022). *ABoVE: Alder shrub cover and soil properties, Alaska, 2019* [Dataset]. ORNL DAAC. https://doi.org/10.3334/ORNLDAAC/2120

PROFESSIONAL PRESENTATION

Talks

Welch, A., Matthews, T., Wang, H., Leong, C., Nielsen, N., Xu, X., Barsanti, K., Sheesley, R., Guenther, A., Czimczik, C. (2023, December). *Characterization of summertime carbonaceous aerosol and biogenic precursors on the North Slope of Alaska.* Oral presentation at the annual American Geophysical Union Meeting, San Francisco, CA, USA.

Posters

Welch, A. M., Nielsen, N., Leong, C., Liu, Y., Wang, H., Teber, D., Guenther, A. B., Bret-Harte, S., Jafarov, E., Czimczik, C. I. (2024, December). *Shrub expansion: Impacts on organic soil and permafrost stability on the North Slope of Alaska*. Poster presented at the annual American Geophysical Union Meeting, Washington, D. C., USA.

Wang, H., Nagalingam, S., **Welch, A. M.,** Leong, C., Czimczik, C. I., Guenther, A. B. (2024, December). *High Temperature Sensitivity of Isoprene Emissions from Sedges: Implications for a Warming Planet*. Poster presented at the annual American Geophysical Union Meeting, Washington, D. C., USA.

Leong, C., Wang, H., **Welch, A**., Nagalingam, S., Czcimzik, C., Guenther, A. (2023, December). *Isoprene emissions of sedge species under varying Arctic climate conditions.* Poster presented at the annual American Geophysical Union Meeting, San Francisco, CA, USA.

Odwuor, A., **Welch, A.,** Li, S., York, R., Banerjee, T., Randerson, J., Czimczik, C. (2023, December). *Large fuel classes as key drivers of PM2.5 emissions from a prescribed fire in California.* Poster presented at the annual American Geophysical Union Meeting, San Francisco, CA, USA.

Sheesley, R., Matthews, T., Moffett, C., **Welch, A.,** Wang, H., Tasnia, A., Guenther, A., Czimczik, C., Barsanti, K. (2023, December). *Biogenic emissions and aerosol response on the North Slope (BEAR-oNS) preliminary 2022 and 2023 results for the North Slope of Alaska: Aerosol composition and realtime VOC.* Poster presented at the annual American Geophysical Union Meeting, San Francisco, CA, USA.

Wang, H., Nagalingam, S., **Welch, A.**, Leong, C., Czimczik, C., Guenther, A. (2023, December). *The pronounced temperature sensitivity of sedge isoprene emission leads to a surge of isoprene release during heatwaves.* Poster presented at the annual American Geophysical Union Meeting, San Francisco, CA, USA.

Matthews, T., Sheesley, R., Moffett., C., Guenther, A., **Welch, A**., Czimczik, C., Wang, H., Barsanti, K., Tasnia, A. (2023, March*). Overview of the BEAR-oNS project and preliminary 2022 results for aerosol analysis from Utqiagvik and Toolik, AK.* Poster presented at the annual South-Central Regional Meeting of the Society of Environmental Toxicology and Chemistry, Denton, TX, USA.

Welch, A., Wang, H., Xu, X., Sheesley, R., Guenther, A., Czimczik, C. (2023, January). *Contribution of ambient biogenic emissions and boreal forest fire smoke to aerosol formation on the North Slope of Alaska.* Poster presented at the biennial Toolik All-Scientist meeting, Santa Barbara, CA, USA.

Wang, H., **Welch, A.,** Nagalingam, S., Brijesh S., Czimczik, C., Guenther, A. (2023, January*). Strong isoprene temperature response differs for woody and herbaceous tundra plants impacting the effect of warming on high-latitude isoprene emission.* Poster presented at the biennial Toolik All-Scientist meeting, Santa Barbara, CA, USA.

Matthews, T., Sheesley, R., Moffett., C., Guenther, A., **Welch, A**., Czimczik, C., Wang, H., Barsanti, K., Tasnia, A. (2023, March*). Overview of the BEAR-oNS project and preliminary 2022 results for aerosol analysis from Utqiagvik and Toolik, AK.* Poster presented at the annual American Meteorological Society Meeting, Houston, TX, USA.

Welch, A., Pedron, S., Jespersen, R., Khazindar, Y., Fiore, N., Goulden, M., Martinez, B., Xu, X., Czimczik, C. (2021, December). *Implications of alder shrub expansion on soil properties in Arctic alpine tundra.* Poster presented at the annual fall meeting of the American Geophysical Union, New Orleans, LA, USA.

Welch, A., Pedron, S., Jespersen, R., Khazindar, Y., Fiore, N., Goulden, M., Martinez, B., Xu, X., Czimczik, C. (2021, November). *Arctic Greening: Implications of expanding shrubs on soil properties in Arctic alpine tundra.* Poster presented at Ridge 2 Reef and UCI Solutions that Scale Joint Poster Session, Irvine, CA, USA.

Welch, A., Dutton, A., & Vyverberg, K. (2019, March). *Evaluation of element/ca ratios as a screening technique to quantify the amount of diagenetic calcite in fossil corals.* Poster presented at the southeastern meeting of the Geological Society of America, Charleston, SC, USA.

EDUCATION/OUTREACH COMMUNICATION

Greco, N. (2024, September) *New Dataset Discussing Aerosols in the Arctic with PhD Candidate* ***Allison Welch***. https://arcticdata.io/all/news/2024/09/new-dataset-discussing-aerosols-in-the-arctic-with-phd-candidate-allison-welch/

Joel, L. (2024, April) *UC Irvine researchers shine light on rapid changes in Arctic and boreal ecosystems*. https://news.uci.edu/2024/04/16/uc-irvine-researchers-shine-light-on-rapid-changes-in-arctic-and-boreal-ecosystems/

Welch, A. (2024, February). *Permafrost in an Impermanent World*. University of California Grad Slam Pitch Competition Semi-Finalist.

Welch, A., Wang, H. (2022, October). *Welcome to Mosquitotown: Fieldwork at Toolik Field Station, AK*. Presentation on summer field work campaign at Toolik Field Station in Arctic Alaska, for the ESS Department's informal seminar series, Half-Baked, run by graduate students for graduate students.

Welch, A., Pahl, B. (2021, November). *Explain Your Research to a Third Grader: The Carbon Cycle*. Science communication outreach content hosted by the University of California, Irvine, School of Physical Sciences, Irvine, CA, USA.

COURSES TAKEN

Technical/Disciplinary
Global Physical Climate
Remote Sensing
Global Biogeochemistry
Climate Data Analysis
Humans in Earth System Science
Paleoclimatology and Paleoceanography
Ecosystem Ecology
Land Surface Processes
Intro to Radiocarbon
Transferable
Data Visualization
Communication Skills
Inclusion & Team Science
Teaching Topics/Pedagogy in Earth System Science
Project Management

SOFTWARE SKILLS

R, Python, ArcGIS/QGIS, Github, Google Earth Engine, Excel, MEGAN, HYSPLIT

PROFESIONAL ASSOCIATIONS

American Geophysical Union